

## Assignment 8: Hybrid Agentic Workflows using LangGraph

### 1. Telecom Domain Assignment

#### Intelligent Network Incident Resolution System

##### Problem Statement

A telecom company receives thousands of network-related complaints daily from customers across multiple regions. Complaints may relate to:

- Slow internet
- Complete outage
- Frequent call drops
- Tower congestion
- Billing-related network disputes
- Device/SIM-related issues

The organization wants to build an **Agentic Incident Resolution System** using LangGraph that can:

1. Understand the complaint
2. Categorize issue severity
3. Check multiple backend systems
4. Dynamically decide escalation paths
5. Generate resolution recommendations
6. Notify appropriate teams/customers

##### Expected Functionalities

##### Inputs

User complaint examples:

- “Internet is extremely slow since morning.”
- “My SIM stopped working after recharge.”
- “Entire area has no signal.”

- “Getting charged despite no connectivity.”

## **Workflow Expectations**

### **Stage 1 — Complaint Understanding**

The system should:

- Extract issue type
- Detect urgency/severity
- Identify whether issue affects:
  - Individual customer
  - Multiple users
  - Entire region

This stage may involve:

- Intent extraction
- Sentiment analysis
- Entity extraction

Learners should think about whether these tasks can run independently.

### **Stage 2 — Backend Investigation**

The system may need to query:

- Tower health systems
- CRM/ticketing systems
- Billing systems
- Device compatibility databases
- Outage monitoring APIs

Some checks may happen simultaneously, while others may depend on earlier classifications.

### **Stage 3 — Intelligent Routing**

Depending on findings:

- Billing issue → Billing Agent
- Tower outage → Network Ops Agent
- Device incompatibility → Device Support Agent
- High-severity outage → Escalation Agent
- Repeated complaints → Priority Handling Agent

Learners must determine:

- Where conditional edges are required
- Which agent receives control dynamically

### **Stage 4 — Resolution Generation**

The system should:

- Summarize findings
- Generate customer-facing response
- Create internal technician notes
- Predict estimated resolution time

Some outputs can be generated simultaneously.

## **2. Healthcare Domain Assignment**

### **Multi-Agent Clinical Triage & Care Coordination System**

#### **Problem Statement**

A hospital wants to automate initial patient triage before assigning doctors. Patients submit symptoms through a portal.

The system should:

- Analyze symptoms
- Assess urgency

- Retrieve medical guidance
- Recommend specialist departments
- Flag emergency conditions
- Coordinate follow-up actions

The workflow should intelligently adapt depending on patient conditions.

## **Expected Functionalities**

### **Inputs**

Examples:

- “Chest pain and shortness of breath.”
- “Fever and headache for 3 days.”
- “Mild skin rash after medication.”
- “Sudden dizziness and blurred vision.”

## **Workflow Expectations**

### **Stage 1 — Symptom Understanding**

System should:

- Extract symptoms
- Detect severity
- Identify possible emergency indicators
- Analyze patient history if available

Potentially independent tasks:

- Symptom extraction
- Risk scoring
- Medical entity recognition
- Sentiment/distress analysis

## **Stage 2 — Knowledge Retrieval**

The workflow may retrieve:

- Relevant medical guidelines
- Drug interaction data
- Specialist recommendations
- Similar historical cases

Learners should decide:

- Which retrievals happen together
- Which depend on severity classification

## **Stage 3 — Intelligent Conditional Routing**

Possible routing:

- Emergency symptoms → Emergency Escalation Agent
- Mild/common symptoms → General Physician Agent
- Medication reactions → Pharmacy Review Agent
- Neurological symptoms → Neuro Specialist Agent
- Cardiac indicators → Cardiology Agent

Some paths may terminate early.

## **Stage 4 — Care Coordination**

The system should:

- Generate triage summary
- Suggest specialist
- Create follow-up instructions
- Recommend diagnostic tests
- Notify emergency desk if needed

Some outputs can execute in parallel.

### **3. Finance Domain Assignment**

#### **Intelligent Loan Risk Assessment & Approval Orchestration System**

##### **Problem Statement**

A financial institution wants to automate loan processing using an agentic system.

The platform should:

- Analyze applicant information
- Verify documents
- Assess fraud risk
- Evaluate creditworthiness
- Dynamically decide approval/escalation/rejection
- Generate personalized loan recommendations

The workflow should adapt based on applicant profile and detected risk.

##### **Expected Functionalities**

###### **Inputs**

Examples:

- Personal details
- Salary slips
- Bank statements
- Loan amount requested
- Employment type
- Credit score

##### **Workflow Expectations**

###### **Stage 1 — Applicant Analysis**

System should:

- Extract applicant profile
- Analyze income stability
- Detect missing documents
- Evaluate spending patterns
- Assess financial behavior

Some analyses may occur simultaneously.

## **Stage 2 — Verification Layer**

Possible checks:

- Credit bureau validation
- Fraud detection
- KYC verification
- Employment verification
- Debt-to-income calculation

Learners must determine:

- Which validations can execute in parallel
- Which require previous outputs

## **Stage 3 — Intelligent Routing**

Possible routing:

- High fraud probability → Fraud Investigation Agent
- Excellent profile → Fast-track Approval Agent
- Missing documents → Clarification Agent
- Medium risk → Manual Review Agent
- High debt ratio → Rejection Recommendation Agent

Conditional logic becomes critical here.

## **Stage 4 — Decision Orchestration**

The system should:

- Generate approval/rejection rationale
- Create risk summary
- Recommend loan amount
- Suggest interest slab
- Notify applicant

Some reporting tasks may happen simultaneously.